

# Acceleration

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

| QUESTION |                                                                             | ANSWER                                                                                                                                 | ✓                        |
|----------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1        | State the standard unit of acceleration.                                    | Metre per second squared ( $\text{m/s}^2$ )                                                                                            | <input type="checkbox"/> |
| 2        | State what the gradient of a velocity–time graph represents.                | Acceleration                                                                                                                           | <input type="checkbox"/> |
| 3        | State what the area under a velocity–time graph represents.                 | The distance travelled                                                                                                                 | <input type="checkbox"/> |
| 4        | State the acceleration of an object in free fall near the Earth's surface.  | About $9.8 \text{ m/s}^2$                                                                                                              | <input type="checkbox"/> |
| 5        | Define terminal velocity.                                                   | The constant velocity of a falling object when the resistive forces equal its weight.                                                  | <input type="checkbox"/> |
| 6        | State the equation linking acceleration, change in velocity and time taken. | acceleration = change in velocity $\div$ time taken<br>( $a = \Delta v \div t$ )                                                       | <input type="checkbox"/> |
| 7        | State the equation for uniform acceleration.                                | (final velocity) <sup>2</sup> – (initial velocity) <sup>2</sup> = 2 $\times$ acceleration $\times$ distance<br>( $v^2 - u^2 = 2 a s$ ) | <input type="checkbox"/> |
| 8        | State what a negative acceleration means.                                   | The object is slowing down.                                                                                                            | <input type="checkbox"/> |
| 9        | State what a horizontal line on a velocity–time graph shows.                | Constant velocity.                                                                                                                     | <input type="checkbox"/> |
| 10       | State what a straight, sloping line on a velocity–time graph shows.         | Constant acceleration.                                                                                                                 | <input type="checkbox"/> |